

Response to Reviewers

Comparing Human and Large Language Model Responses to Patients' Online Questions: Towards Multi-dimensional Patient-centered Support (Application #14169)

We thank the reviewers for their careful reading and constructive feedback. Below we address each point and indicate the corresponding changes in the manuscript.

Methods

1. Dataset curation

Comment. The methods for curation of the dataset could be clearer. The cited dataset included more than 300 forum questions across forums on four conditions. A description of the source dataset and of how it was reduced to 122 would help readers understand the methods and generalizability. (Reviewer 1, Meta Review; Reviewer 3, Main Round)

Response. We expanded the dataset description to document the full provenance and the inclusion criterion. The revised passage reads:

The foundation of our study was a curated dataset of 122 English-language question posts and their threaded replies (N=519 comments). They were derived from a prior study that manually reviewed a random sample of 1,000 posts, itself drawn from approximately 60,000 posts flagged as containing laboratory test results (for example, a number followed by units such as 10 u/mL) and confirmed 321 posts that contained the poster's laboratory test results and one or more questions. These 321 posts spanned roughly 50 condition-specific forums on the online health community, concentrated in thyroid disorders, hepatitis, heart disease, and HIV. For the present study, we retained the 122 posts that had threaded replies, which we collected from the original forum threads, since the prior study analyzed the questions only.

2. Codebook

Comment. The qualitative coding framework lacks sufficient detail. Operational definitions for categories such as low, mid, and high emotional support are needed to address reproducibility and subjectivity. (Reviewer 1, Meta Review; Reviewer 2, Main Round)

Response. We added operational definitions for the scored dimensions. For emotional support, the anchors are:

Low: little or no emotional support; the response is largely informational with no acknowledgment of the poster's feelings. Mid: some emotional support; brief or generic encouragement or acknowledgment. High: substantial emotional support; explicit empathy, validation of the specific situation, and personalized reassurance.

3. Prompt engineering and reproducibility of prompts

Comment. Prompt engineering may be needed; without a good prompt the full potential of the model is not exploited. (Reviewer 2). A brief appendix or table showing one or two examples of the prompt variations the authors observed would increase reproducibility. (Reviewer 1, Main Round)

Response. Our use of the original user-posted questions as fixed prompts was a deliberate design choice. We study how laypeople query widely available tools in situ, not the upper bound of model capability under expert prompt optimization. Pilot testing showed that small changes in wording and structure produced meaningful differences in responses, which motivated fixing the prompts to the original questions for consistency and a fair comparison across the four models and against peer responses on identical inputs.

We agree that prompt-optimized performance is an important complementary question and note it as future work.

Variant A (original post).

I recently had thyroid levels tested because of numerous hypothyroid symptoms. The results are: TSH: 1.8, Free T4: 0.79 L (0.8-1.7), Free T3: 244 (210-440). Negative on antibodies. My question is, with a low Free T4 should my TSH be higher? My doctor mentioned Central Hypothyroidism and to retest in 4 weeks. What are your thoughts about my levels along with hypo symptoms?

Model (GPT 4o) response to Variant A.

Your test results and symptoms suggest a complex situation that needs careful evaluation. Normally, when Free T4 and Free T3 are low, the pituitary releases more TSH to stimulate the thyroid. In primary hypothyroidism you would typically see a high TSH alongside low Free T4. However, your TSH is normal and your Free T4 is slightly below range. This could suggest central (secondary) hypothyroidism, where the problem lies with the pituitary rather than the thyroid; in that case TSH can be low or normal despite low thyroid hormone. Your doctor's plan to retest in a few weeks is sensible, and assessing other pituitary hormones may help. Since you have hypothyroid symptoms, raise these concerns with your doctor, who may consider treatment if symptoms persist. Keep monitoring how you feel and communicate any changes to your provider, who can tailor advice to your situation.

Variant B (same case restructured).

Are these thyroid results normal? TSH 1.8, Free T4 0.79 (range 0.8-1.7), Free T3 244 (range 210-440), thyroid antibodies negative. With a low Free T4, shouldn't my TSH be higher?

Model (GPT 4o) response to Variant B.

Your labs are mostly normal except for a very slightly low Free T4. You're correct that in classic primary hypothyroidism, a low Free T4 should usually raise TSH. Since your TSH is normal at 1.8, this pattern is not typical primary hypothyroidism. Common explanations:

- Normal lab variation. Your Free T4 is barely below range.
- Temporary suppression from illness, stress, low calories, poor sleep, or medications or supplements.
- Less commonly, central hypothyroidism, where TSH stays normal despite low thyroid hormone.

Your normal Free T3 and negative antibodies are reassuring. Most doctors would:

- Repeat TSH and Free T4 in 6 to 12 weeks.
- Review symptoms and medications or supplements.
- Investigate further only if Free T4 stays low or symptoms are significant.

By itself, this pattern is borderline, not clearly abnormal.

Observed difference. Holding the clinical content and the model constant and changing only the prompt produced a different response. The differential emphasis shifted. Variant A carried the poster's own mention of central hypothyroidism, and the model foregrounded central (secondary) hypothyroidism as the leading explanation and discussed further pituitary testing. Variant B removed that cue, and the model listed central hypothyroidism last, under "less commonly," behind normal laboratory variation and temporary suppression. The overall framing also shifted: Variant A treated the result as a complex situation requiring careful evaluation, while Variant B concluded the pattern was borderline and not clearly abnormal.

4. Writing and clarity

Comment. The phrase "about of stomach flu" should read "a bout of stomach flu" in the personalization to individual needs section. (Reviewer 3, Main Round)

Response. Thank you for catching this. We corrected the phrase to "a bout of stomach flu" in the personalization to individual needs section and reread the surrounding text for similar errors.

Discussion

5. Disclosure of AI involvement in online health communities

Comment. A more detailed recommendation on how online health communities might disclose AI involvement would strengthen the discussion. (Reviewer 1, Main Round)

Response. We expanded the recommendation. The added text reads:

Building on the diabetes community's stance toward AI, we recommend that online health communities incorporating AI clearly label AI-generated and AI-assisted replies, make such assistance opt-in, and let each community set its own disclosure norms rather than having them imposed by the platform.

Limitations

6. General-purpose models only

Comment. The evaluation is restricted to general-purpose models and does not include domain-specific medical models, limiting generalizability. (Reviewer 2, Main Round)

Response. We added this point to the first limitation. The revised passage reads:

We evaluated only general-purpose LLMs and no medically fine-tuned or domain-specific models. This reflects our focus on the widely available tools patients' access, since specialized systems remain out of reach for most laypeople due to cost or familiarity. Domain-specific models may perform differently and benchmarking them against general-purpose tools is an important direction for future work.